

Temporal Multiple Rotation Averaging: A Multi-Camera Dataset

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Abstract—This paper presents a comprehensive time-series image dataset tailored to evaluate multi-target tracking within a dynamic sensor network using multiple rotation averaging techniques. Recognizing the significant gap in the availability of datasets—particularly those offering details on camera orientations and positions—our dataset stands out by comprising images from multiple vehicles equipped with cameras navigating through a common environment. Each image in the series is supplemented with exact ground truth values for the respective camera's orientation and location at every discrete time step. Such a rich dataset is a vital resource for techniques aiming to deduce camera positions and orientations and finds relevance in target tracking and information fusion tasks. Going beyond mere introduction, we leverage this dataset to showcase the superiority of our newly proposed temporal multiple-rotating averaging method. When juxtaposed with prevailing non-temporal techniques, our method manifests a markedly enhanced accuracy, underlining its potential for real-world applications.

Index Terms—multiple rotation averaging, structure from motion, 3D rotation, temporal information, particle filter, multi-target tracking

I. INTRODUCTION

As the Internet of Things (IoT) continues its rapid expansion and the accessibility of cost-effective vision sensors grows, camera networks are witnessing a surge in adoption across a diverse array of surveillance applications. Not only are these networks integral to multi-object tracking [1], but they also play pivotal roles in the intricacies of autonomous vehicle control [2], [3], [4], facilitating smoother navigation and safer decision-making. Additionally, they are becoming indispensable in tasks like multi-view reconstruction [5], enabling comprehensive environmental mappings and more immersive virtual experiences. The rising prominence of these camera networks underscores their versatility and the immense potential to shape the next phase of technological evolution in surveillance and beyond.

Precise knowledge of camera locations is paramount to harnessing the capabilities of camera networks for various applications such as mapping, path planning, control, target tracking, and information fusion. In scenarios where cameras remain static, a one-time localization suffices. However, continuous estimation of their locations becomes essential in dynamic settings, like cameras mounted on moving vehicles. Common techniques to ascertain sensor positioning often combine GPS with IMU sensors or rely on visual odometry. GPS and IMU sensors can accurately measure position and

orientation, respectively, in ideal scenarios [6]; however, they are expensive and can fail in situations without a satellite signal (e.g. in tunnels or remote locations). Visual odometry is commonly used for UAVs [7]; in scenarios involving high speed, it may not be accurate, and it relies on solving scale ambiguity [8].

Multiple Rotation Averaging (MRA) is a method for estimating a set of camera orientations, using just the visual information captured by the cameras [9]. MRA is the problem of using a set of noisy relative rotation measurements between image pairs to estimate the absolute rotations of the cameras. MRA is a part of the Structure-from-Motion (SfM) problem that has been used successfully in scenarios including image reconstruction [10] and estimating camera orientations [11].

MRA can be solved using several approaches. The most common technique is non-linear optimization. Traditionally, bundle adjustment methods were used [12], which are computationally expensive and require an accurate initial estimate. Quicker iterative non-optimal approaches were developed that took advantage of the underlying Lie algebra structure of 3D rotations [13]. Further improvements through advances in areas including alternative loss functions [14], [10] and optimization methods [15]. Low-rank and sparse matrix decomposition is a non-iterative approach introduced by Candès et al. [16] that reduces the dimensionality of the data while preserving as much information as possible [17], [18]. Recently, artificial neural networks have also been used for MRA, initially using the neural network to calculate the weights for each relative rotation [19], further methods were developed that use a message passing neural network (MPNN) to simulate the structure of the camera network [20], [21], [22].

Recently, a new approach to MRA was proposed that utilizes the temporal information available in the problem where the cameras take a sequence of images, and their location is dynamically changing through time [23]. However, this method was only tested on synthetic datasets, rather than real camera data. In this paper, real camera datasets were created to validate the proposed MRA approach. Figure 1 illustrates an example of the images captured in these datasets. This paper's contributions are:

- Enhancing the distributed temporal MRA method as initially proposed by Blair et al. 2023 [23];
- Developing comprehensive datasets based on authentic camera data, specifically designed for rigorous testing and

validation of MRA techniques.

The rest of the paper is structured as follows. In Section II we provide an overview of the mathematical and theoretical background behind the proposed MRA method. In Section III, we introduce the novel contributions of the paper. Section IV explains the experimental setup, presents the experimental results and discusses the findings. Section V concludes the paper.

II. BACKGROUND

3D rotations, fundamental in many fields such as computer graphics, robotics, and computer vision, can be represented in a myriad of formats. These representations are crucial for performing operations and transformations in three-dimensional spaces. These include:

- the 3×3 orthonormal rotation matrix representation $R \in \mathbb{R}^{3 \times 3}$, i.e. $R^\top R = I_3$ and $\det(R) = 1$,
- the Euler angle representation (also known as yaw, pitch and roll), which is three successive rotations of $\theta = (\alpha, \beta, \gamma)$ in degrees around specified rotation axes,
- the axis-angle representation $\mathbf{v} = \theta \hat{\mathbf{v}} = [v_1 \ v_2 \ v_3]^\top$ that is a single rotation θ around axis $\hat{\mathbf{v}}$, and
- the quaternion representation $\mathbf{q} = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, where $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{i}\mathbf{j}\mathbf{k} = -1$ and $\mathbf{q}^2 = -1$.

Each of these formats has its own advantages and challenges, and the choice often depends on the specific application and computational requirements in question.

The same rotation can be easily transformed from any of these representations to any of the other representations. The elements of a rotation matrix corresponding to the Euler angles (α, β, γ) are given by

$$\begin{aligned} \text{row 1: } R_{11} &= \cos(\beta) \cos(\gamma) \\ R_{12} &= \sin(\alpha) \sin(\beta) \cos(\gamma) - \cos(\alpha) \sin(\gamma) \\ R_{13} &= \cos(\alpha) \sin(\beta) \cos(\gamma) + \sin(\alpha) \sin(\gamma) \\ \text{row 2: } R_{21} &= \cos(\beta) \sin(\gamma) \\ R_{22} &= \sin(\alpha) \sin(\beta) \sin(\gamma) + \cos(\alpha) \cos(\gamma) \\ R_{23} &= \cos(\alpha) \sin(\beta) \sin(\gamma) - \sin(\alpha) \cos(\gamma) \\ \text{row 3: } R_{31} &= -\sin(\beta) \\ R_{32} &= \sin(\alpha) \cos(\beta) \\ R_{33} &= \cos(\alpha) \cos(\beta) \end{aligned} \quad (1)$$

Given the rotation matrix, Euler angles are given by:

$$\begin{aligned} \alpha &= \tan^{-1}(R_{32}/R_{33}) \\ \beta &= -\sin^{-1}(R_{31}) \\ \gamma &= \cos^{-1}\left(R_{11}/\sqrt{1-R_{31}^2}\right). \end{aligned} \quad (2)$$

3D rotation matrices belong to the Special Orthogonal group $\mathbf{SO}(3)$, $R \in \mathbf{SO}(3)$, which are related to the equivalent axis-angle representation $\mathbf{v}$ through the associated Lie algebra $[\mathbf{v}]_\times \in \mathfrak{so}(3)$, where $[\mathbf{v}]_\times$ is the skew-symmetric form of $\mathbf{v}$

$$\mathbf{\Omega} = [\mathbf{v}]_\times = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix}. \quad (3)$$

The transformation between the rotation matrix representation and the axis-angle representation is given by the matrix exponential and matrix logarithm functions,

$$R = \exp(\mathbf{\Omega}) \quad (4)$$

$$\mathbf{\Omega} = \log(R) \quad (5)$$

which are defined by

$$\exp(A) = I + \sum_{k=1}^{\infty} \frac{A^k}{k!} \quad (6)$$

$$\log(A) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{(A - I)^k}{k} \quad (7)$$

Further information on Lie groups Lie algebras and the matrix exponential and logarithm functions can be found here [24]. The transformations between the quaternion representation $\mathbf{q}$ and the angle-axis representation $\mathbf{v}$ are given by

$$\mathbf{q} = [\cos(\theta), \hat{\mathbf{v}}\sin(\theta)] \quad (8)$$

and

$$\begin{aligned} \theta &= 2 \arccos(w) \\ \mathbf{v} &= [x \ y \ z]/\sqrt{1-w^2} \end{aligned} \quad (9)$$

If the absolute rotation of cameras i and n in a given reference frame are $R^{(i)}$ and $R^{(n)}$, then the relative rotation from n to i is given by

$$R^{(n,i)} = R^{(i)} R^{(n)-1}. \quad (10)$$

MRA is a subset of the Structure-from-Motion problem. It involves the process of estimating the 3D structure of an object or scene from multiple 2D images captured from different viewpoints. MRA utilizes mathematical algorithms to triangulate the features of an object or scene to reconstruct its shape and size accurately. In the context of this paper, the aim is to find the absolute rotations of a set of cameras, given a collection of noisy pairwise relative rotation measurements, in addition to the absolute orientation of a single specified camera (known as the reference camera). The network of cameras can be described by the graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where the vertices $\mathcal{V}$ represent the cameras and the edges $\mathcal{E}$ represent the relative relationship between those cameras. The number of vertices/cameras is denoted by $|\mathcal{V}| = N$ and the number of edges by $|\mathcal{E}| = M$.

Representing a camera, each node is characterized by its label and the absolute rotation of the camera, which is denoted by $R^{(i)}$ for node $(i) \in \mathcal{V}$. Each edge in the graph corresponds to a communication link between two cameras through which the images are sent and received and is characterized by a pair comprised of the node labels and a weight that represents the relative rotation between the two cameras. The weight associated with the edge $(n, i) \in \mathcal{E}$ is denoted by $R^{(n,i)}$ and is assumed to be calculated via feature matching between overlapping images acquired by the two cameras.

MRA is the task of finding the absolute rotations, $\{R^{(i)}\}_{i=1}^N$, of each node in the network, given a set of noisy relative

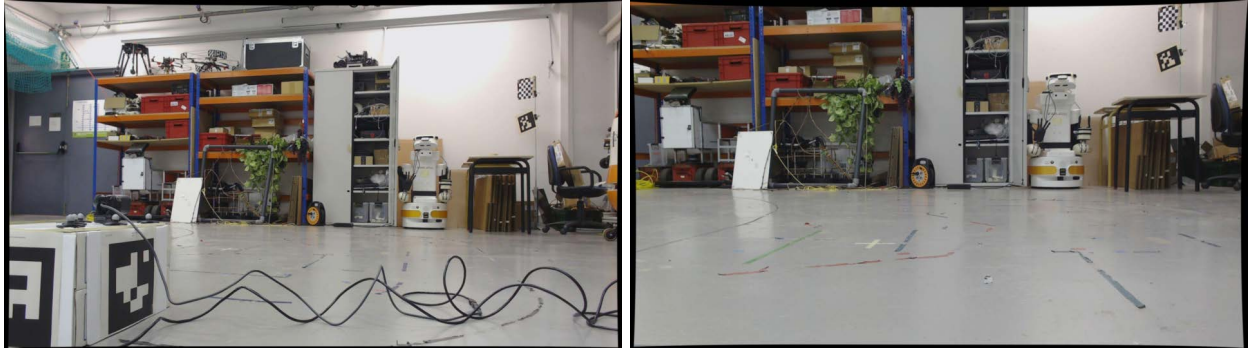


Fig. 1: Two images from two different cameras, captured at the same time step.

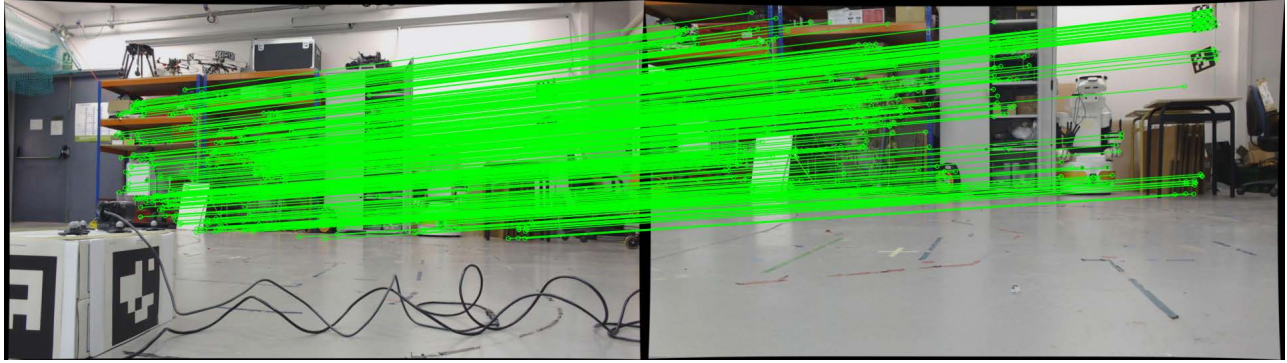


Fig. 2: Two images from two different cameras at the same time step. The matched feature points are marked by green circles and are connected between the images by green lines.

camera orientations $\{\tilde{R}^{(n,i)}\}$ and a reference camera rotation. With no loss of generality, the reference camera index is assumed to be $i = 1$, and its orientation to be $\theta_0^{(1)} = \mathbf{0}$. The most common approach to solving the MRA problem is via finding a solution for the following optimization problem [15]:

$$R^{(i)*} = \arg \min_{\{R^{(i)}\}} \sum_{(n,i) \in \mathcal{E}} \rho \left(d(\tilde{R}^{(n,i)}, R^{(i)} R^{(n)^{-1}}) \right) \quad (11)$$

where $\rho(\cdot)$ is a loss function such as the ℓ_1 or ℓ_2 loss functions [10] and $d(\cdot)$ is a distance measure between two rotations, e.g. the geodesic, chordal or quaternion distance [9]. While there can be up to $C_2^N = N(N-1)/2$ relative rotations in the network, a minimum of $M = N-1$ suitable edges are required to fully span the network and allow for solving the optimization problem of (11). In practice, there are usually $|\mathcal{E}| > N-1$ relative rotations, providing some redundant information to improve accuracy.

For further details on the mathematical background of rotations, rotation distance metrics and rotation averaging, see Hartley et al. [9].

III. PROPOSED METHOD

The proposed method can be separated into two components: First, the relative rotation measurements between image pairs need to be calculated. Second, the camera orientations need to be estimated through MRA.

The relative rotations between image pairs are calculated using the following method. First, the images are undistorted using the camera's intrinsic parameters. Then feature keypoints and descriptors are extracted from each image using the SIFT (Scale-Invariant Feature Transform) algorithm [25]. SIFT was found to result in better matches than the alternative detectors that were tried such as ORB (Oriented FAST and rotated BRIEF) [26]. For each image pair in the same time step, their features are matched using the FLANN (Fast Library for Approximate Nearest Neighbors) algorithm [27]. The best feature matches are kept and used to find the essential matrix relating the two sets of keypoints. An example of the matched keypoints from two images is shown in Figure 2. This essential matrix is then decomposed into the rotation and translation elements. This rotation matrix describes the relative rotation between the two images.

Figure 3 illustrates the accuracy of the feature-matching algorithm. Images taken from two separate cameras at the same time step are processed, features are extracted and matched, the homography matrix relating the first camera's position to the second camera's position is calculated from these keypoints and used to warp the first image to match the perspective of the second camera.

This paper builds on the recently proposed temporal MRA approach [23] to estimate the absolute camera orientations. The method has been extended to include more than one

previous time step in the set of temporal relative rotation measurements.

In scenarios with sequential image data (e.g. video) where each camera records an image at each time step, using a greater number of measurements by including relative rotations from a previous time step to the current time step is expected to result in a more accurate estimation. These additional relative rotations are referred to as *temporal relative rotations* [23]. In previous work, this was limited to including relative rotations between one camera i from the current time step k and one camera n from the previous time step, $R_{k-1,k}^{(n,i)}$. However, this can be generalized to include all possible image pairs across multiple time steps. Let us denote two arbitrary time steps that are before or at the current time $k_1 \leq k$ and $k_2 \leq k$, where $k_2 \geq k_1$. Camera i 's orientation at time k_1 is represented by the rotation matrix $R_{k_1}^{(i)}$ and camera n 's orientation at time k_2 is represented by the rotation matrix $R_{k_2}^{(n)}$. Therefore the relative rotation from camera n at time k_2 to camera i at time k_1 by $R_{k_2,k_1}^{(n,i)}$.

This paper proposes using all possible relative rotation measurements of camera pairs across a sequence of time steps from a specified number of time steps before the current time step $k - N_k$, up to the current time step k . The set of all possible relative rotation measurements given these constraints is

$$\{R_{k_2,k_1}^{(n,i)} | n, i \in \mathcal{V}, k_1, k_2 \in [k - N_k, k]\} \quad (12)$$

In practice not all of these possible relative rotations will have measurements, for example, $R_{k_1,k_1}^{(i,i)}$ will just be the identity rotation and not useful, and there may be cases where for particular values of i , n , k_1 and k_2 , $R_{k_2,k_1}^{(n,i)}$ is not able to be accurately calculated from the proposed method described above. This adjustment to the restrictions on the set of possible relative rotations allows for significantly more measurements to be included, leading to more potential information and more robustness to measurement noise and outliers.

This proposed MRA algorithm will be referred to herein as the Extended Temporal MRA algorithm.

IV. EXPERIMENT AND RESULTS

A. Experiment

The experimental setup for creating the datasets is described below. Six vehicles were simulated with boxes with a front-facing camera attached to the top. Attached to the top surfaces of these boxes are five marker balls, to allow for the orientation and position of the boxes to be accurately measured in real-time using the OptiTrack system [28]. Additionally, each of the four walls of the boxes is marked with a unique Aruco tag [29]. These can be used for detecting the boxes for further applications such as target tracking.

The locations of the vehicles were restricted to a 5 metre by 5 metre square. The position of each vehicle was measured relative to one of the OptiTrack marker balls, designated as the lead marker. For each vehicle, the offset distance from the lead marker to the camera lens was measured, and the difference in orientation between the camera orientation and

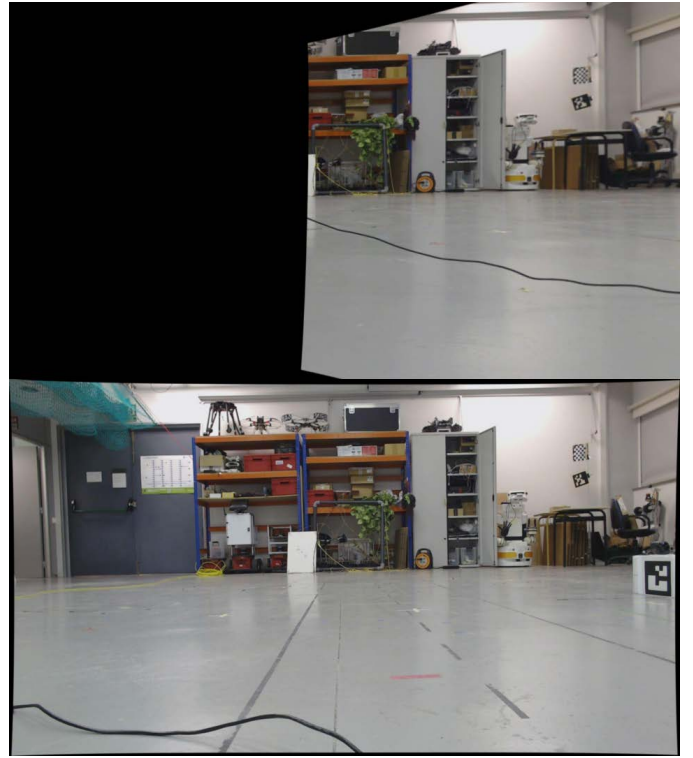


Fig. 3: The top frame is the image from the first of two cameras, warped to the perspective of the second camera, based on the homography matrix relating to the camera pair. The bottom frame is the original image from the second camera.

the measured box orientation was also measured. This ensures that an accurate ground truth of the cameras' positions and orientations is recorded.

Vehicle driving scenarios were simulated by moving the vehicles along noisy curvilinear trajectories. Each vehicle was moved a small distance along its trajectory, then each camera took an image and the orientation and translation data was read and saved from the OptiTrack system. The OptiTrack data and images were recorded using a custom ROS (Robot Operating System) script, to ensure sufficient synchronization between the cameras and the OptiTrack system. Through taking these sequential discrete "snapshots" we created "stop-motion" scenarios that simulate a continuous scenario with data recorded at periodic time intervals. Each scenario was run for 40 time steps. Figure 4 illustrates the vehicle trajectories and associated camera FoVs (Fields of View) for one of the scenarios.

Each dataset comprises three parts: The camera data, the OptiTrack data and the metadata. The camera data consists of the sets of RGB images taken by each camera at each time step. The OptiTrack data consists of the 3D translation (in cartesian coordinates) and orientation (in quaternion form) of each vehicle at each time step. The metadata consists of the miscellaneous data that is necessary for image processing, OptiTrack data processing etc., including the extrinsic camera

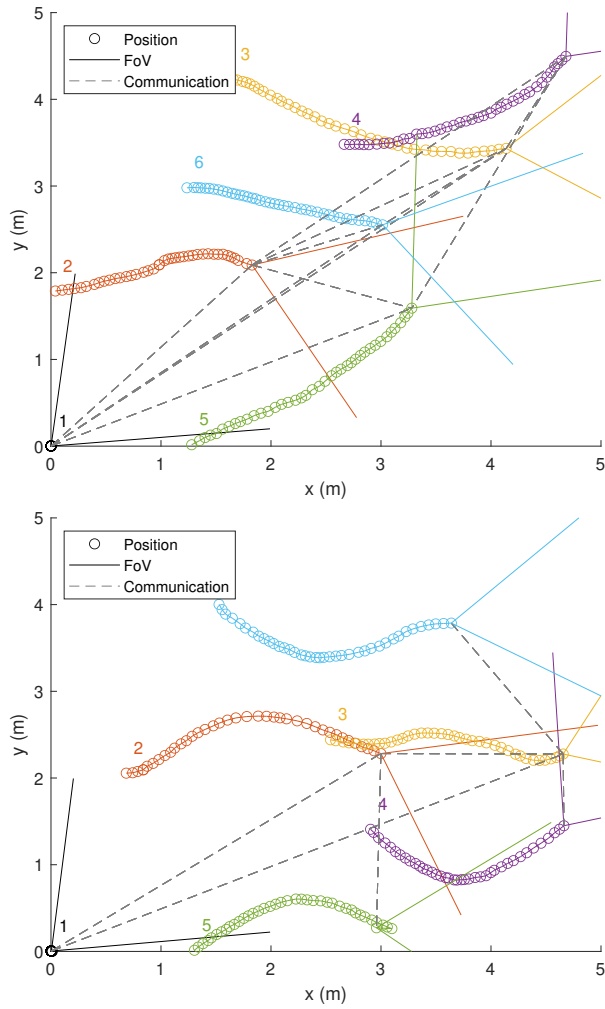


Fig. 4: The top-down view of vehicle trajectories in the 1st (top) and 2nd (bottom) scenarios. The position of each vehicle is shown at each time step by a circle. The solid lines denote the FoVs of the front-facing cameras and the dotted lines denote sensor communication (at the final time step). Note the stationary position of camera 1 at the origin of the figures.

parameters (offset distances and angles from the lead markers to the cameras) for creating ground truth values and the intrinsic camera parameters (focal length, principal point offset and skew) for undistorting the images. The dataset is publicly available for download as a Github repository[30].

Two scenarios from the dataset were used to test the proposed Extended Temporal MRA method, the first being relatively simple (shown in Figure 4) and the second being more complex. The Extended Temporal MRA method was compared to both the original Temporal MRA method [23] and a baseline method, the 1DSfM algorithm proposed by Wilson & Snavely [31], which does not incorporate any temporal information, i.e. each time step is treated as an isolated MRA problem. While the 1DSfM method produces the same results each run, both of the Temporal MRA methods have a small

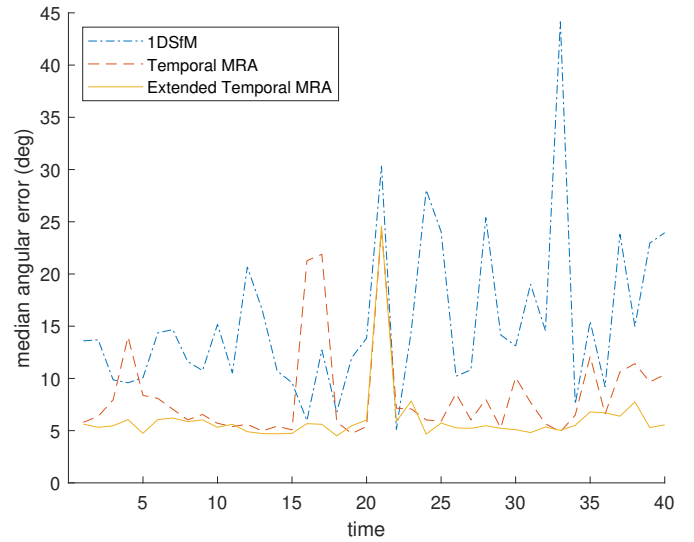


Fig. 5: Experimental results on the first scenario in the dataset. The median angular error was calculated at each time step after each MRA algorithm was performed.

variance in the output of each run, so they were run 10 times and the results were averaged. For the Extended Temporal MRA method, the number of time steps before the current time step that is included for creating temporal relative rotations i $N_k = 2$. Increasing N_k beyond this was found to not yield significant improvements in accuracy while increasing computation time.

B. Results

Figures 5 & 6 compare the accuracy of the proposed Extended Temporal MRA method against both the Temporal MRA method and the baseline method, by comparing the median angular error of the cameras, where the lower the angular error, the more accurate the orientation estimates. In the first scenario both Temporal MRA methods significantly outperformed the baseline method and the Extended Temporal MRA method improved upon the original, with an average angular error across all time steps of 5.5905° , compared to Temporal MRA's average angular error of 6.9532° and the baseline's average angular error of 15.3555° . These results are replicated in the more challenging second scenario, with the Temporal MRA methods showing a greater improvement over the baseline methods and the Extended Temporal MRA method again improving upon the original, with an average angular error across all time steps of 5.4080° , compared to Temporal MRA's average angular error of 5.7167° and the baseline's average angular error of 21.9699° .

V. CONCLUSION

In this work, a dataset comprising several scenarios of a distributed network of vehicles with attached cameras was created. Additionally, the recently proposed temporal multiple rotation averaging algorithm was extended and improved. This multiple rotation averaging algorithm is tested on the created

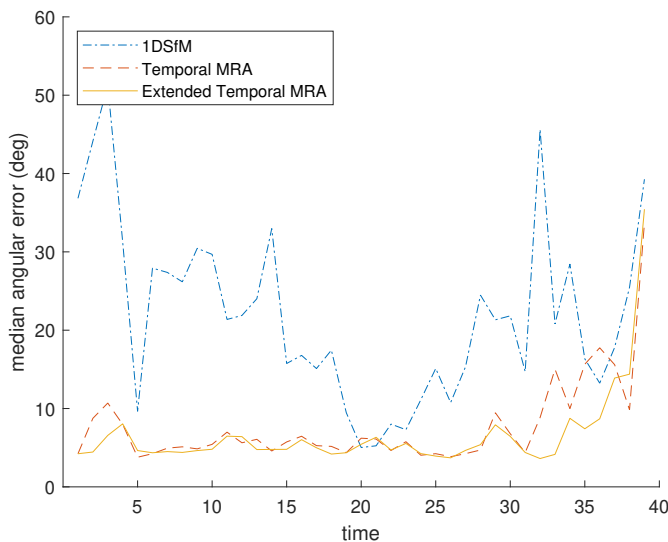


Fig. 6: Experimental results on the second scenario in the dataset. The median angular error was calculated at each time step after each MRA algorithm was performed.

dataset and demonstrates significantly improved accuracy over the baseline algorithm and the original temporal multiple rotation averaging algorithm. Future work can be done to incorporate camera translations to achieve full vehicle pose estimation. Additionally, although not tested in this paper, the dataset is capable of being used for testing multi-target tracking algorithms.

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